

What is the clinical significance of TP53 Mutation in Lung Adenocarcinoma in CIViC?

Understood! I'll research the clinical significance of TP53 mutations in lung adenocarcinoma as reported in the CIViC (Clinical Interpretations of Variants in Cancer) database. I'll provide you with the latest information and relevant citations.

Worked for 3 minutes >

In the CIViC knowledge base, the molecular profile "**TP53 Mutation**" is classified as a **prognostic** variant in lung adenocarcinoma. CIViC curators have summarized two evidence items for this variant:

1. **Reduced progression-free survival** – An evidence item (EID6032) based on the study by Imielinski et al. (2012) examined 135 chemotherapy-naïve lung adenocarcinoma patients. Patients with tumors harboring **TP53 mutations** showed **significantly shorter progression-free survival** compared with those lacking the mutation ($p = 0.0014$, log-rank test) [civcdb.org](#). CIViC classifies this evidence as **level B (clinical)**, assigns the evidence type "Prognostic," and labels the **clinical significance as "poor outcome"** [civcdb.org](#).
2. **Association with poor survival after resection** – A second evidence item (EID6031) drawn from Kosaka et al. (2009) analyzed 321 patients with lung adenocarcinoma who underwent potentially curative pulmonary resection. Tumors with **TP53 mutations (exons 4–10)** were associated with **significantly poorer overall survival** than wild-type tumors (log-rank $P = 0.023$; $HR \approx 1.98$) [civcdb.org](#). However, this association lost significance after adjustment in a multivariate Cox regression [civcdb.org](#). This evidence item is also classified as level B, with the type "Prognostic" and **significance "poor outcome"** [civcdb.org](#).

Clinical significance in CIViC: Combining these evidence items, CIViC interprets **TP53 mutation** in lung adenocarcinoma as a **poor prognostic marker**. Tumors with somatic TP53 mutations are associated with worse progression-free or overall survival compared with TP53 wild-type tumors [civcdb.org](#) [civcdb.org](#). Thus, within CIViC, the clinical significance of TP53 mutation in lung adenocarcinoma is "poor outcome," indicating that patients with this mutation tend to have a worse prognosis.

Sources

Describe a task

+ | Agent x Sources v

